

Titanic2_Yuliia_Chernysheva

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1 Titanic Data Analysis Project (Part2)

This second stage of the analysis builds upon the initial Exploratory Data Analysis (EDA) of the Titanic dataset. While Part 1 focused on age-based survival and financial overview, Part 2 investigates the intersection of gender and social class to understand how these factors dictated survival outcomes.

Research Questions

1. **Question 1:** What is the distribution of passengers lost vs. survived by **sex**?
2. **Question 2:** What is the distribution of passengers lost vs. survived by **ticket class**?
3. **Question 3:** What is the **conditional probability** of survival given a person's sex and passenger class?

2 Data Inspection and Preparation

2.1 Selection and Verification of Data

To ensure the integrity of the analysis, the dataset is re-inspected. This step documents the available variables and verifies if any missing values (NaN) could affect the calculations for sex, class, or survival.

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Constants for the project colors
COLOR_LOST = '#E63946'
COLOR_SURVIVED = '#0077B6'
COLOR_TEXT = '#2F2F2F'
COLOR_GRID = '#E0E0E0'

# Load Data
df = pd.read_csv('titanic.csv')
```

```

# Documenting columns and checking for missing values
print(">> Dataset Columns:")
print(df.columns.tolist())

print("\n>> Missing Values Count:")
print(df.isnull().sum())

# Data Summary
print("\n>> Data Summary:")
display(df.describe())

```

```

>> Dataset Columns:
['Survived', 'Pclass', 'Name', 'Sex', 'Age', 'Siblings/Spouses Aboard',
'Parents/Children Aboard', 'Fare']

```

```

>> Missing Values Count:

```

```

Survived          0
Pclass            0
Name              0
Sex               0
Age               0
Siblings/Spouses Aboard  0
Parents/Children Aboard  0
Fare              0
dtype: int64

```

```

>> Data Summary:

```

	Survived	Pclass	Age	Siblings/Spouses Aboard \
count	887.000000	887.000000	887.000000	887.000000
mean	0.385569	2.305524	29.471443	0.525366
std	0.487004	0.836662	14.121908	1.104669
min	0.000000	1.000000	0.420000	0.000000
25%	0.000000	2.000000	20.250000	0.000000
50%	0.000000	3.000000	28.000000	0.000000
75%	1.000000	3.000000	38.000000	1.000000
max	1.000000	3.000000	80.000000	8.000000

	Parents/Children Aboard	Fare
count	887.000000	887.000000
mean	0.383315	32.30542
std	0.807466	49.78204
min	0.000000	0.000000
25%	0.000000	7.92500
50%	0.000000	14.45420
75%	0.000000	31.13750
max	6.000000	512.32920

2.2 Technical Summary

- **Data Integrity:** The dataset is complete, with no missing values in the **Sex**, **Pclass**, or **Survived** columns. This allows for a precise calculation of survival probabilities without data cleaning.
- **Variable Types:** The analysis will utilize categorical data (**Sex** and **Class**) to group passengers and numerical data (**Survived**) to measure outcomes.
- **Ready for Analysis:** With the data structure verified, the investigation can now proceed to analyze how gender and social standing directly influenced survival rates.

3 Research Question 1: Survival by Sex

Question: What is the distribution of passengers lost vs. survived based on their gender?

To answer this, the data is grouped by the *Sex* and *Survived* columns. This transformation allows us to see the exact count of outcomes for both men and women.

```
[2]: # Grouping by Sex and Survival
sex_survival = df.groupby(['Sex', 'Survived']).size().unstack()
sex_survival.columns = ['Lost', 'Survived']

print(">> Survival Count by Sex:")
display(sex_survival)

# Visualization: Bar Chart
chart_colors = [COLOR_LOST, COLOR_SURVIVED]

ax = sex_survival.plot(kind='bar', figsize=(10,6), color=chart_colors,
                      width=0.8, edgecolor=COLOR_GRID)

# Add Percentage
for i, gender in enumerate(sex_survival.index):
    # two columns for each sex: Died (0) and Survived (1)
    for j, survived_status in enumerate([0, 1]):
        # Locate the specific patch(bar)
        # Indexing in patches works as follows
        p = ax.patches[j * len(sex_survival) + i]
        height = p.get_height()
        total_in_class = sex_survival.loc[gender].sum()
        percentage = f' {(height/total_in_class):.1%}'

        ax.annotate(percentage, (p.get_x() + p.get_width()/2, height + 5),
                    ha='center', va='bottom', fontsize=11,
                    fontweight='bold', color=COLOR_TEXT)

# Settings
plt.title('Lost vs Survived by Sex', color=COLOR_TEXT, fontweight='bold',
         fontsize=18, pad=20)
```

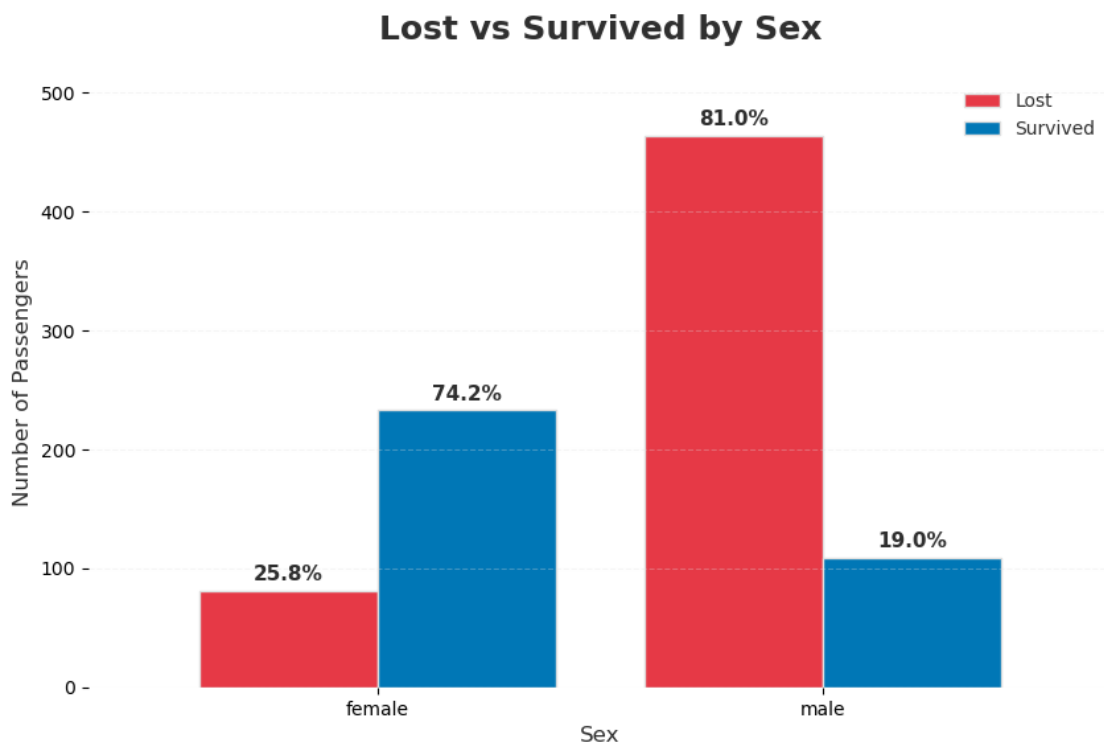
```
plt.xlabel('Sex', color=COLOR_TEXT, fontsize=12)
plt.ylabel('Number of Passengers', color=COLOR_TEXT, fontsize=12)
plt.legend(labelcolor=COLOR_TEXT, frameon=False)
plt.xticks(rotation=0)
plt.grid(axis='y', color=COLOR_GRID, linestyle='--', alpha=0.3)
plt.box(False)

# Increase the limit on Y
plt.ylim(0, sex_survival.values.max() + 50)

plt.show()
```

>> Survival Count by Sex:

	Lost	Survived
Sex		
female	81	233
male	464	109



Analysis of Question 1:

Stark Contrast: The data reveals a massive survival gap between genders. Female passengers had a survival rate of over 70%, while the majority of males were lost.

Conclusion: This statistically supports the historical “women and children first” policy, making

Sex one of the strongest predictors of survival in the Titanic disaster.

4 Research Question 2: Survival by Class

Question: How did the passenger class affect the number of people who were lost or survived?

To analyze the impact of socio-economic status, the dataset is grouped by *Pclass* and *Survived*. This transformation aggregates individual records into a summary of outcomes for each of the three ticket classes.

```
[3]: # Group by Pclass and Survival
class_survival = df.groupby(['Pclass', 'Survived']).size().unstack()
class_survival.columns = ['Lost', 'Survived']

print(">> Survival Count by Ticket Class:")
display(class_survival)

ax = class_survival.plot(kind='bar', figsize=(10,6), color=chart_colors,
                        width=0.8, edgecolor=COLOR_GRID)

# Add Percentage
for i, pclass in enumerate(class_survival.index):
    # two columns for each class: Died (0) and Survived (1)
    for j, survived_status in enumerate([0, 1]):
        # Locate the specific patch(bar)
        # Indexing in patches works as follows
        p = ax.patches[j * len(class_survival) + i]
        height = p.get_height()
        total_in_class = class_survival.loc[pclass].sum()
        percentage = f'{(height/total_in_class):.1%}'

        ax.annotate(percentage, (p.get_x() + p.get_width()/2, height + 5),
                    ha='center', va='bottom', fontsize=11,
                    fontweight='bold', color=COLOR_TEXT)

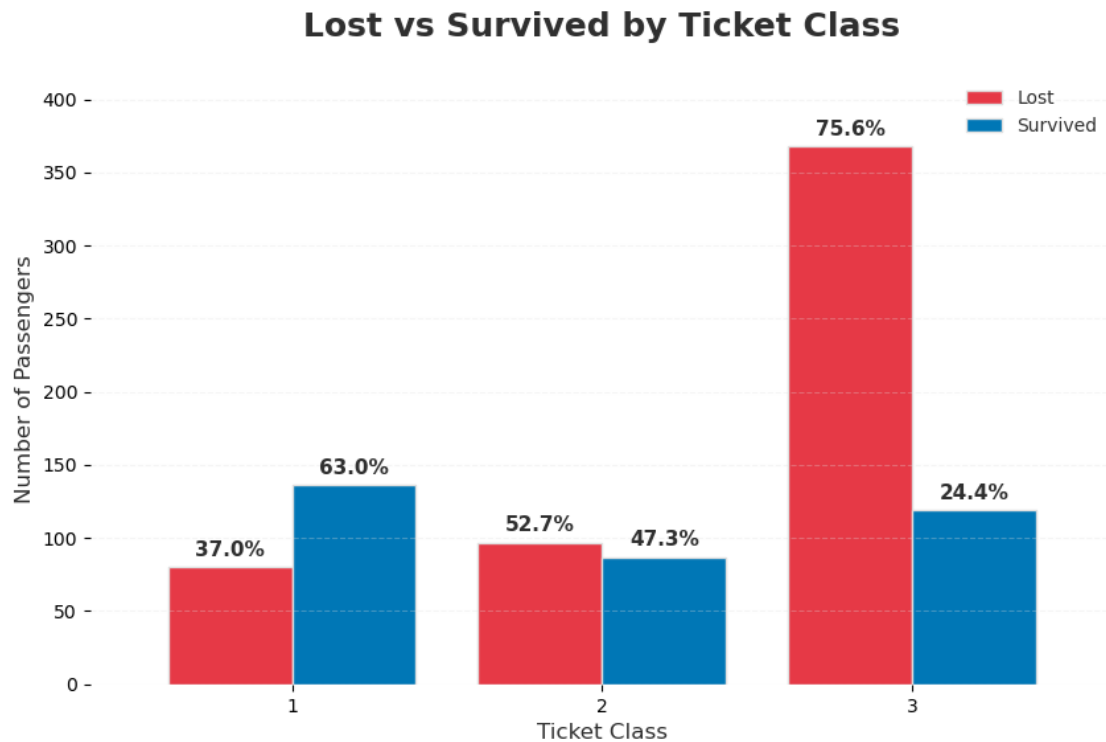
# Settings
plt.title('Lost vs Survived by Ticket Class', color=COLOR_TEXT,
         fontweight='bold', fontsize=18, pad=20)
plt.xlabel('Ticket Class', color=COLOR_TEXT, fontsize=12)
plt.ylabel('Number of Passengers', color=COLOR_TEXT, fontsize=12)
plt.legend(labelcolor=COLOR_TEXT, frameon=False)
plt.xticks(rotation=0)
plt.grid(axis='y', color=COLOR_GRID, linestyle='--', alpha=0.3)
plt.box(False)

# Increase the limit on Y
plt.ylim(0, class_survival.values.max() + 50)
```

```
plt.show()
```

>> Survival Count by Ticket Class:

Pclass	Lost	Survived
1	80	136
2	97	87
3	368	119



Analysis Summary:

- **First Class Advantage:** Passengers in 1st class had the highest survival rate (**63.0%**), representing the only group where survivors outnumbered casualties.
- **The Struggle of the 3rd Class:** In contrast, 3rd class passengers faced a tragic mortality rate, with only **24.4%** survival and over 350 lives lost.
- **The Middle Ground:** 2nd class survival was nearly a 50/50 split, at **47.3%**.

Conclusion: Social standing (Pclass) was a critical determinant of survival, often overriding other demographic factors during the disaster.

5 Research Question 3: Conditional Probability of Survival

Question: What is the conditional probability that a person survives given their sex and passenger-class?

To find the probability for each specific group (e.g., a female in 1st class), we calculate the **mean** of the Survived column after grouping by both Sex and Pclass. Since Survived is coded as 0 and 1, the average value directly represents the probability of survival.

```
[4]: # Conditional probability: P(Survival | Sex, Class)
# Group by Sex and Pclass
# Calculate sum (survivors), count (total), and mean (probability)
stats = df.groupby(['Sex', 'Pclass'])['Survived'].agg(
    survived_count='sum',
    passengers='count',
    survival_prob='mean'
).reset_index()

# Create a text percentage for display
stats['survival_prob_pct'] = (stats['survival_prob'] * 100).round(1).
    ↪astype(str) + '%'

# Rename columns before outputting
final_table = stats.rename(columns={
    'Sex': 'Sex',
    'Pclass': 'Ticket Class',
    'passengers': 'Total Passengers',
    'survived_count': 'Survivors',
    'survival_prob_pct': 'Survival Rate (%)'
})[['Sex', 'Ticket Class', 'Total Passengers', 'Survivors', 'Survival Rate_
    ↪(%)']]

print(">> Conditional Survival Analysis (Summary Table):")
display(final_table)

# Visualization: Grouped Bar Chart for Probabilities
chart_data = stats.pivot(index='Sex', columns='Pclass',
    values='survival_prob') * 100
ax = chart_data.plot(kind='bar', figsize=(10, 6),
    color=['#A8DADC', '#457B9D', '#1D3557'],
    edgecolor=COLOR_GRID, width=0.8)

# Adding labels on top of bars
for p in ax.patches:
    ax.annotate(f'{p.get_height():.1f}%',
        (p.get_x() + p.get_width() / 2, p.get_height() + 2),
        ha='center', va='center', fontsize=11,
        fontweight='bold', color=COLOR_TEXT)

plt.title('Survival Probability by Sex and Class', color=COLOR_TEXT,
    fontweight='bold', fontsize=18, pad=20)
plt.xlabel('Sex', color=COLOR_TEXT, fontsize=12)
```

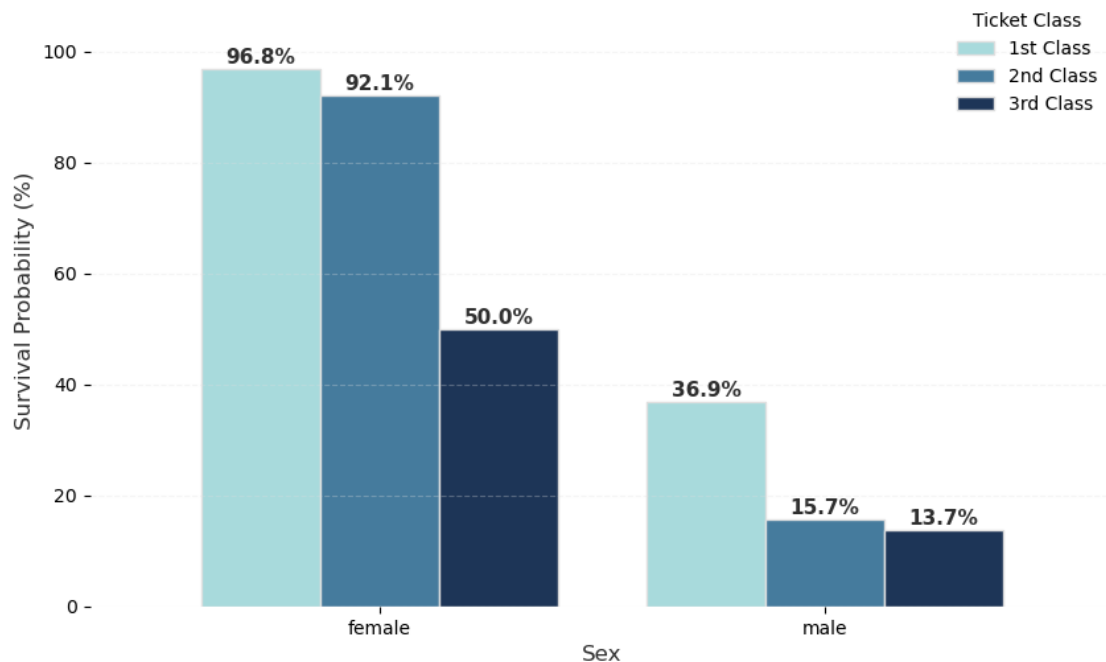
```
plt.ylabel('Survival Probability (%)', color=COLOR_TEXT, fontsize=12)
plt.legend(['1st Class', '2nd Class', '3rd Class'], title='Ticket Class',
           frameon=False)

plt.xticks(rotation=0)
plt.grid(axis='y', color=COLOR_GRID, linestyle='--', alpha=0.3)
plt.box(False)
plt.ylim(0,110) # Probability doesn't exceed 100%
plt.show()
```

>> Conditional Survival Analysis (Summary Table):

	Sex	Ticket Class	Total Passengers	Survivors	Survival Rate (%)
0	female	1	94	91	96.8%
1	female	2	76	70	92.1%
2	female	3	144	72	50.0%
3	male	1	122	45	36.9%
4	male	2	108	17	15.7%
5	male	3	343	47	13.7%

Survival Probability by Sex and Class



Analysis of Results

- **The “Women and Children First” Reality:** The probability for females in 1st Class was the highest at 96.8%, and 92.1% for 2nd Class. This shows that being female in a higher class almost guaranteed survival.

- **The 3rd Class Divide:** Even for women, being in 3rd Class dropped the survival probability significantly to 50.0%.
- **Male Mortality:** Men in all classes faced low probabilities, but the drop-off is extreme: while 36.9% of 1st-class men survived, only 13.7% of 3rd-class men did.

Conclusion: The conditional probability proves that while gender was the primary factor, the passenger class acted as a secondary filter that drastically altered one's chances of survival.

6 Final Conclusion

The final analysis of the Titanic dataset demonstrates that passenger survival was largely dictated by social status and gender, reflecting the rigid maritime traditions and class hierarchies of the era.

Key Findings:

- **The Gender Factor:** Biological sex was the primary determinant of survival. Due to the “women and children first” protocol, the survival rate for females exceeded **70%**, while the vast majority of male passengers were lost.
- **Class Inequality:** Passenger class acted as a critical secondary filter. Those with higher-class tickets had better access to the boat deck, resulting in a **63.0%** survival rate in 1st Class compared to only **24.4%** in 3rd Class.
- **The Combined Effect:** The conditional probability analysis highlights how class influenced chances even within the same gender. For instance, women in 3rd Class had a **50.0%** survival rate, significantly lower than their counterparts in 1st and 2nd Class.
- **Male Survival Trends:** Even 1st Class status did not guarantee safety for men; their survival probability was **36.9%**, though this was still nearly three times higher than that of men in 3rd Class (**13.7%**).

Summary:

The data confirms that an individual's life expectancy on the night of the disaster was not a matter of luck. The combination of gender and ticket price effectively predetermined the fate of most passengers, making the Titanic disaster a clear historical example of socio-economic inequality.

[]: